

Literature Status for arXiv:math/0210228: L_p and Partition–Weight Norms

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Status

This is a `literature_already_answered` packet. It records that one question from the final open-problems section of Alspach–Tong, *Subspaces of L_p , $p > 2$, Determined by Partitions and Weights*, arXiv:math/0210228, was answered by the later Alspach–Tong note *Subspaces of L_p , $p > 2$, with unconditional bases have equivalent partition and weight norms*, arXiv:math/0310343.

Source Question

In the “Remarks and Open Problems” section of arXiv:math/0210228, Alspach and Tong ask whether L_p is isomorphic to a space whose norm is given by partitions and weights. This question occurs after their discussion of the Bourgain–Rosenthal–Schechtman complemented subspaces and before their question about constants in a Rosenthal-type inequality. The source paper is throughout concerned with the range $2 < p < \infty$.

Supporting Answer

The abstract of arXiv:math/0310343 states that every subspace of L_p , $2 < p < \infty$, with an unconditional basis has an equivalent norm determined by partitions and weights, and then explicitly says that consequently L_p has such a norm.

The body of the paper gives the precise theorem:

if $X \subset L_p$, $2 < p < \infty$, has an unconditional basis, then X has an equivalent partition–weight norm.

The immediately following corollary states:

$L_p[0, 1]$, $2 < p < \infty$, has an equivalent norm given by partitions and weights.

Since $L_p[0, 1]$ has an unconditional Haar basis for $1 < p < \infty$, this corollary gives an affirmative answer, up to isomorphism, to the source question in the $p > 2$ setting of arXiv:math/0210228.

Scope

This packet is intentionally narrow. It does not claim answers to the other questions in the same open-problems section: existence of an envelope-property example not embedding into L_p , necessary and sufficient conditions for partition–weight spaces to embed into L_p , the tensor-product question, the transfinite X_p^α versus R_p^α question, or sharp constants in the displayed Rosenthal-type inequality.

Search Evidence

The lane-6 sparse queue selected arXiv:math/0210228. Cheap run indexes were searched for the arXiv id, title words, **partitions and weights**, **envelope property**, X_p^α , R_p^α , and Rosenthal keywords; no duplicate packet for this source was found. A local full-source search for exact phrases from the open-problems section found arXiv:math/0310343. Its abstract, theorem, and corollary directly identify the affirmative answer to the L_p question.

References

- [1] D. E. Alspach and S. Tong, *Subspaces of L_p , $p > 2$, Determined by Partitions and Weights*, arXiv:math/0210228.
- [2] D. E. Alspach and S. Tong, *Subspaces of L_p , $p > 2$, with unconditional bases have equivalent partition and weight norms*, arXiv:math/0310343.